

Executive Summary

13577252 Canada Inc., operating as DMcL Consulting (DMcL) was retained by Impala Canada Ltd. | Lac des Iles (LDI) to prepare an up to date Emission Summary and Dispersion Modelling (ESDM) report for their LDI Mine facility (the Facility) located north northeast of Thunder Bay, Ontario. This updated ESDM report has been prepared in support of an application to amend their existing facility's Environmental Compliance Approval (ECA) (ECA Number No. 6122-AC9NNJ, Issued September 13th, 2016). A copy of the current ECA is included in Appendix A.

LDI operates a palladium ore mine and processing mill on site to produce a palladium rich concentrate which is shipped off site for further processing and refining. The facility is located approximately 13 km West of Highway 527, 90 km north of Thunder Bay, Ontario and is located on land leased from the Ministry Of Northern Development, Mines, Natural Resources and Forestry (MNDM). The site consists of a total of 74 mining claims including 418 claim units (see figure C-5 in appendix C for the claim map).

Significant operations at the Facility include the operation of:

- the existing and expanded open pit mine,
- the underground mine complex,
- underground and surface raw ore crushing,
- stationary and semi-mobile grinding and stacking equipment and
- a Mill Complex consisting of a number of (wet) fine grinding mills, gravity settling and concentrating processes.

Supporting Facilities on-site include:

- an operations camp,
- tailings and water management facilities,
- ore and waste rock stockpiles,
- maintenance and repair shops,
- a network of unpaved roads and
- the metallurgical Assay Lab.

The Facility is connected to the local power grid, however there are three (3) large emergency diesel power generations on site to provide power to essential services in case of emergency power outages.

Updates to the ESDM report have been prepared to address the proposed increase in the production limit for the Mill's operations from the current: 15,000 tonnes of ore processed per day, to 30,000 tonnes of ore processed per day. This includes potential increases in emissions from vehicular traffic on facility roads, ore crushing and screening operations, and the associated assay lab sample processing.

Other updates incorporated in this revision (rev.12) include the input of significant Facility buildings to the dispersion model to better account for potential building downwash effects, clarification of and existing material handling operations, the addition of a number of material handling operations not previously captured, minor source location updates as well as significant updates to the network of surface haul roads. These updates have been documented in the Facility's Modification Log - As per Condition 5.1 of the ECA.

This ESDM report has been prepared according to the requirements as spelled out in the MECP guidance document "*Procedure for Preparing an Emissions Summary Dispersion Modelling Report, Version 4.0, February 2017 [Guideline A-10]*" (ESDM Guide).

The Source Summary Tables presenting the Facility's air emission sources and the associated emission rate estimates are presented as Table B1 and Table B1a in Appendix B of this report.

Advanced air dispersion modelling of the Facility's air emissions was performed according to the MECP guidance document "*Air Dispersion Modelling Guideline for Ontario, version 4.0, February 2017*" (ADMGO); using the Lakes Environmental Inc. AERMOD View (ver.10.2.1) and the updated AERMOD and AERMET version 19191 dispersion modelling software. The maximum Point of Impingement (POI) concentrations for contaminants of concern were determined based on the aggregate worst case contaminant emission rates from the Facility.

The Facility's NAICS code is 212299 (Metal Ore Mining) and as such, is listed under Schedule 4 of O.Reg. 419/05; requiring it to comply with the Point of Impingement (POI) standards set out in Schedule 3 of the regulation. The maximum modelled concentrations of all contaminants of concern have been compared to the POI limits and guidelines set forth in the MECP's Air Contaminant Benchmarks (ACB) List (April 2018) to determine compliance with the relevant regulatory standards.

The modeled results, for all contaminants of concern, indicate that the maximum POI concentrations off site are expected to be less than the respective POI limits. This ESDM report demonstrates that the LDI Facility is capable of operating in compliance with O.Reg. 419/05 for all contaminants of concern under maximum, worst case operating scenarios.

An Acoustic Assessment Report has not been prepared for the Facility as the site meets the minimum separation distance from the nearest sensitive receptor (>1km) as per the attached Section 9 Noise Screening form in Appendix F.

The Emission Summary Table below (also provided as Table B2 and Table B2a in Appendix B), shows the comparison of modelled results against the relevant regulatory standard and demonstrates that LDI is capable of operating in compliance with all applicable POI limits for all contaminants of concern.

Table 1: Emission Summary Table – Comparison of Modelling Results with Schedule 3 POI Standards

Contaminant of Concern	CAS Number	Aggregate Emission Rate (g/s)	Maximum Ground Level POI Concentration (µg/m³)			Regulatory Criterion (µg/m³)			% of Criterion		
			1 hr	24 hr	Other	1 hr	24 hr	Other	1 hr	24 hr	Other
Carbon Monoxide	630-08-0	11.4			114(1/2hr)			6000 (1/2hr)			2% (1/2hr)
Nitrogen Oxides	10102-44-0	4.4 (1 hr)	40			400			10%		
		2.1 (24 hr)		3			200			1.7%	
		21.3 (1/2 hr)			165			1880			9%
Lead	7439-92-1	0.0075		0.0088	0.0019 (Month)		0.5	0.2 (Month)		1.8%	0.9% (Month)
TSP Total	N/A	133.2		104	21 (Annual)		120			86%	
Speciated Metals Contained in Road Dust Emissions¹											
Antimony-Sb	440-36-0	0.00002		0.0000			25		-	0.0%	-
Arsenic-As	440-38-2	0.000133	0.0003	0.00010		1	0.3		0.0%	0.0%	-
Barium-Ba	440-39-3	0.0096	0.018	0.0075		30	10		0.1%	0.1%	-
Cadmium-Cd	7440-43-9	0.0000133		0.000010			0.25		-	0.0%	-
Chromium –Cr	7440-47-3	0.0255		0.0199			0.5			4.0%	-
Cobalt-Co	7440-48-4	0.0120	0.023	0.0093		0.3	0.1		7.6%	9.3%	
Copper-Cu	7440-50-8	0.110		0.0856			50		-	0.2%	-
Magnesium-Mg	1309-48-4	15.72		12			120		-	10.2%	-
Manganese-Mn	7439-96-5	0.136		0.106			2.5		-	4.2%	-
Mercury-Hg	7439-97-6	0.00002		0.00002			2		-	0.0%	-
Molybdenum-Mo	7439-98-7	0.000143		0.000111			120		-	0.0%	-
Nickel-Ni	7440-02-0	0.155		0.120	0.0243		2	0.04 Annual*	-	6.0%	61%
Palladium-Pd	10/1/7657	0.00040		0.0003			10		-	0.0%	-

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			1 hr	24 hr	Other	1 hr	24 hr	Other	1 hr	24 hr	Other
Platinum-Pt	6/4/7440	0.000031		0.00002			0.2		-	0.0%	-
Selenium-Se	7782-49-2	0.000133		0.00010			10		-	0.0%	-
Silver-Ag	7440-22-4	0.000021		0.00002			1		-	0.0%	
Strontium-Sr	1314-11-0	0.0262		0.0204			120		-	0.0%	-
Titanium-Ti	7440-32-6	0.119		0.093			120		-	0.1%	-
Vanadium-V	7440-62-2	0.0140		0.0109			2		-	0.5%	-
Zinc-Zn	7440-66-6	0.00562		0.0044			120		-	0.0%	-

* Annual Assessment Value - to be compared against the worst case annual average POI concentration

¹ Calculated from POI Concentration of Total Suspended Particulate x (%) Composition of each metal species in Ore